

**a****Natural-language prompt**

"Design a CT scanner  
for clinical brain imaging"

sole user input

PWM  
Agent

**spec.md (excerpt)**

```
modality: ct
carrier: x-ray
primitives:
  - Project (Radon, 180 angles)
  - Convolve (detector blur)
  - Poisson (1e5 photons)
solver: FBP + TV ( $\lambda=0.02$ )
gate_1: rank(A)/N = 0.89
gate_2: SNR = 42.1 dB
gate_3:  $\Delta\text{CoR} < 0.5$  px
```

**b****Triad Gate Certificate****Gate 1: Recoverability****PASS**

rank(A)/N = 0.89 > 0.5  
Information capacity  
sufficient for reconstruction

**Gate 2: Carrier Budget****PASS**

SNR = 42.1 dB > 20 dB  
Photon flux adequate  
for target resolution

**Gate 3: Op. Mismatch****PASS**

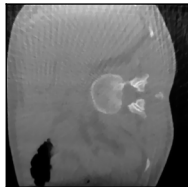
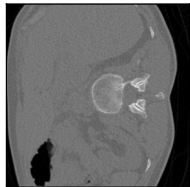
$\Delta\text{CoR} = 0.3$  px < 0.5 px  
Calibration within  
tolerance bounds

**c**

Ground truth

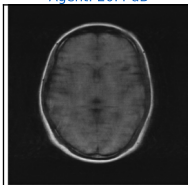
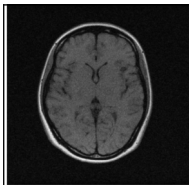
Reconstruction

CT



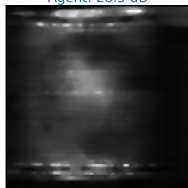
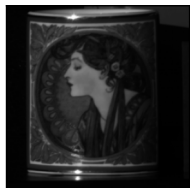
Agent: 20.4 dB

MRI



Agent: 28.5 dB

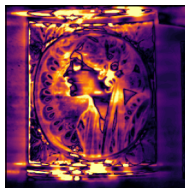
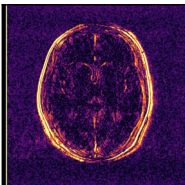
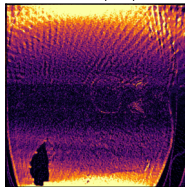
CASSI



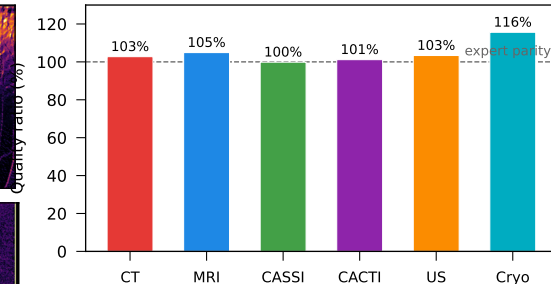
Agent: 25.4 dB

**d**

Error (x5)



Quality ratio (%)

**Quality ratio (agent / expert)****Theorem 1 tightness ratio**